

Total marks – 80**Attempt Questions 1–5**

Answer each question in a SEPARATE writing booklet. Extra writing booklets are available.

Question 1 (14 Marks) Use a SEPARATE writing booklet **Marks**

- (a) For the curve. $y = 4 + 3x - x^3$,
- (i) find any stationary points and determine their nature. **4**
 - (ii) find any points of inflexion. **2**
 - (iii) find the co-ordinates of the y-intercept. **1**
 - (iv) sketch the curve in the domain $-3 \leq x \leq 3$ showing all the above features. **2**
- (b) The sum of the radii of two circles is 100 cm. If one of the circles has a radius of x cm
- (i) Show that the sum of the areas of the two circles is given by **2**
$$A = 2\pi(x^2 - 100x + 5000).$$
 - (ii) Find the least possible value for this area. **3**

Question 2 (14 Marks) Use a SEPARATE writing booklet

- (a) Find the centre and radius of this circle $x^2 + 14x + 14 + y^2 - 2y = 0$. **3**
- (b) Express $5x^2 + 2x - 3$ in the form $A(x+1)^2 + B(x+1) + C$. **3**
- (c) For what values of k does the equation $kx^2 - 4kx - (k-5) = 0$ have real roots? **3**

Question 2 continues on Page 2.

Question 2 Continued.**Marks**

- (d) The vertex of the parabola is $(-7, -2)$ and the focus is $(-3, -2)$. Find
- (i) the equation of the directrix. 1
 - (ii) the equation of the parabola. 1
- (e) Derive the equation of the locus of the point $P(x, y)$ that moves so that it is equidistant from the point $A(-6, 5)$ and the point $B(3, -1)$. 3

Question 3 (16 Marks) Use a SEPARATE writing booklet

- (a) Find:
- (i) $\int (4x^3 + 7x^2 - 3) dx$. 1
 - (ii) $\int \frac{6x^3 - 7x}{x^2} dx$. 2
 - (iii) $\int x\sqrt{x} dx$ 2
- (b) Evaluate $\int_0^2 (2x - 1)^3 dx$. 2
- (c) Find the equation of the curve that passes through the point $(2, 5)$, given that the gradient function is $3 + 2x - x^2$. 3
- (d) (i) Sketch, on the same axes $y = 10 - x^2$ and $y = x + 4$. 2
- (ii) Hence find the area bounded by parabola $y = 10 - x^2$ and the line $y = x + 4$. 4

Question 4 (16 Marks) Use a SEPARATE writing booklet**Marks**

(a) Differentiate

(i) e^{4x+8} 1

(ii) $\frac{e^x}{e^x + 1}$ 2

(b) Evaluate $\int_0^2 e^{3x} dx$. 2

(c) Find $\int 8x^3 e^{x^4} dx$. 2

(d) Find the equation of the tangent to the curve $y = e^{x^2-1}$ at $x = 1$. 3

(e) Find the exact volume of the solid of revolution formed when the curve 3

$y = e^x - e^{-x}$ is rotated about the x -axis between $x = 0$ and $x = \frac{1}{2}$.

(f) Use Simpson's Rule with five function values to find an approximation 3

to $\int_0^4 e^{-x^2} dx$, correct to two decimal places.

Question 5 (20 Marks) Use a SEPARATE writing booklet(a) Find $\frac{dy}{dx}$ given

(i) $y = \log_e(2x + 7)$ 1

(ii) $y = \log_e(2x + 1)(x - 5)$ 2

(iii) $y = x^3 \log_e x$ 2

(b) Evaluate $\int_2^3 \frac{8x}{2x^2 + 7} dx$. 2

Question 5 continues on Page 4.

Question 5 Continued.**Marks**

- (c) (i) Show $\frac{4x+3}{2x+1} = 2 + \frac{1}{2x+1}$. **1**
- (ii) Hence find $\int \frac{4x+3}{2x+1} dx$ **2**
- (d) Simplify $2\log_3 6 + \log_3 18 - 3\log_3 2$. **3**
- (e) Solve
- (i) $\log_2 64 = x$ **1**
- (ii) $3^x = 5$, correct to two decimal places. **2**
- (f) (i) Sketch the curve $y = \log_e x$. **1**
- (ii) Find the area enclosed between the curve $y = \log_e x$, the x axis and the line $x = 2$. Shade this area on your diagram. **3**

END OF THE PAPER

Question 1

a) $y = 4 + 3x - x^3$

$$\frac{dy}{dx} = 3 - 3x^2$$

$$\frac{d^2y}{dx^2} = -6x$$

For stationary points $\frac{dy}{dx} = 0$

$$3 - 3x^2 = 0$$

$$3(1 - x^2) = 0$$

$$3(1 - x)(1 + x) = 0$$

$$x = \pm 1$$

When $x = 1$

$$y = 4 + 3x - x^3$$

$$= 4 + 3(1) - (1)^3$$

$$= 6$$

$\therefore (1, 6)$ is a stationary point.

Test the nature

$$\frac{d^2y}{dx^2} = -6x \text{ at } x = 1$$

$$= -6(1)$$

$$= -6$$

As $\frac{d^2y}{dx^2} < 0$ the curve is concave down $\curvearrowright$

$\therefore (1, 6)$ is a local maximum

When $x = -1$

$$y = 4 + 3x - x^3$$

$$= 4 + 3(-1) - (-1)^3$$

$$= 2$$

$\therefore (-1, 2)$ is a stationary point.

Test the nature

$$\frac{d^2y}{dx^2} = -6x \text{ at } x = -1$$

$$= -6(-1)$$

$$= 6$$

As $\frac{d^2y}{dx^2} > 0$ the curve is concave up

$\therefore (-1, 2)$ is a local minimum.

ii) For points of inflexion

$$\frac{d^2y}{dx^2} = 0 \text{ and there is a}$$

change in concavity

$$\frac{d^2y}{dx^2} = -6x$$

$$-6x = 0$$

$$x = 0$$

$$y = 4 + 3x - x^3 \text{ at } x = 0$$

$$= 4 + 3(0) - 0$$

$$= 4$$

$\therefore (0, 4)$ may be a point of inflexion

check concavity

x	0^-	0	0^+
$\frac{d^2y}{dx^2} = -6x$	+		-

$\therefore$ There is a change in concavity

$\therefore (0, 4)$ is a point of inflexion

(iii) the y intercept is $(0, 4)$

(iv) Endpoints

$$\begin{aligned} \text{when } x = -3, y &= 4 + 3x - x^3 \\ &= 4 + 3(-3) - (-3)^3 \\ &= 22 \end{aligned}$$

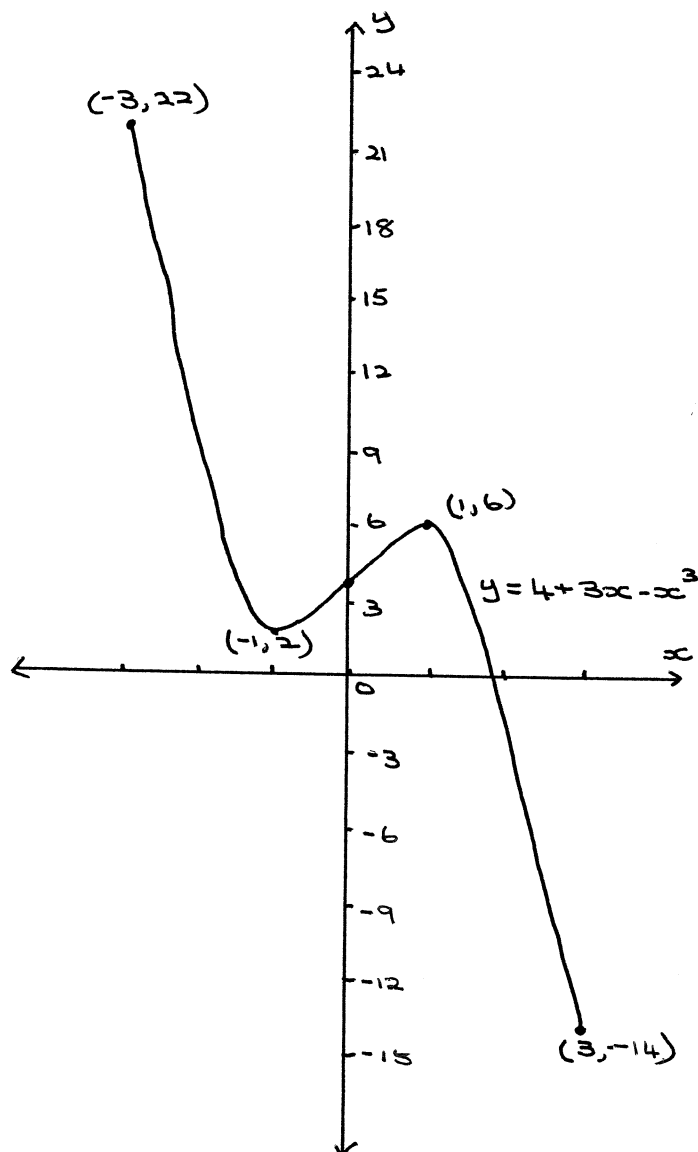
when $x=3$

$$y = 4 + 3x - x^3$$

$$= 4 + 3(3) - 3^3$$

$$= -14$$

$\therefore$ The endpoints are $(-3, 22)$ and $(3, -14)$



b) radius of 1 circle = x
radius of the other circle = $100-x$

$$A = \pi r^2 + \pi R^2$$

$$= \pi x^2 + \pi (100-x)^2$$

$$= \pi x^2 + 10000\pi - 200\pi x + \pi x^2$$

$$= 2\pi x^2 - 200\pi x + 10000\pi$$

$$\therefore A = 2\pi (x^2 - 100x + 5000)$$

$$1) A = 2\pi x^2 - 200\pi x + 10000\pi$$

$$\frac{dA}{dx} = 4\pi x - 200\pi$$

$$\frac{d^2A}{dx^2} = 4\pi$$

For stationary points $\frac{dA}{dx} = 0$

$$\therefore 4\pi x - 200\pi = 0$$

$$4\pi x = 200\pi$$

$$x = 50$$

when $x=50$

$$A = 2\pi x^2 - 200\pi x + 10000\pi$$

$$= 2\pi (50)^2 - 200\pi (50) + 10000\pi$$

$$= 5000\pi$$

As $\frac{d^2A}{dx^2} > 0$, a minimum value occurs

$\therefore$ The least possible area is $5000\pi \text{ cm}^2$

Question 2

$$a) x^2 + 14x + 14 + y^2 - 2y = 0$$

$$x^2 + 14x + \left(\frac{14}{2}\right)^2 + y^2 - 2y + \left(\frac{-2}{2}\right)^2 = -14 + \left(\frac{14}{2}\right)^2 + \left(\frac{-2}{2}\right)^2$$

$$(x+7)^2 + (y-1)^2 = -14 + 49 + 1$$

$$(x+7)^2 + (y-1)^2 = 36$$

$$(x+7)^2 + (y-1)^2 = 6^2$$

$\therefore$ centre is $(-7, 1)$

radius is 6

$$b) 5x^2 + 2x - 3$$

$$= A(x+1)^2 + B(x+1) + C$$

$$= Ax^2 + 2Ax + A + Bx + B + C$$

$$= Ax^2 + (2A+B)x + (A+B+C)$$

$$\therefore A = 5 \quad 2A+B = 2 \quad A+B+C = -3$$

$$2(5)+B = 2 \quad 5-8+C = -3$$

$$B = -8 \quad C = 0$$

$$\therefore 5x^2 + 2x - 3 \equiv 5(x+1)^2 - 8(x+1)$$

$$c) kx^2 - 4kx - (k-5) = 0$$

$$\Delta = b^2 - 4ac$$

$$= (-4k)^2 - 4(k) \times -(k-5)$$

$$= 16k^2 + 4k^2 - 20k$$

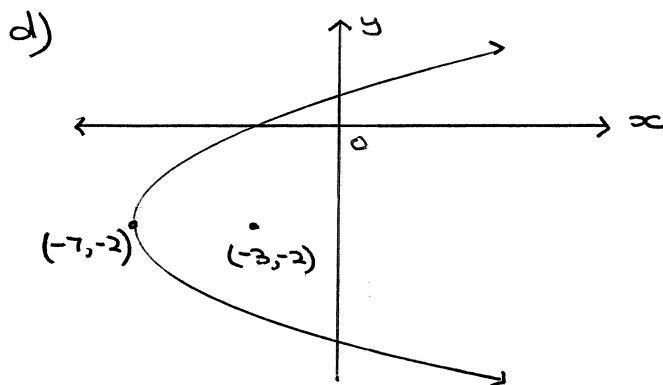
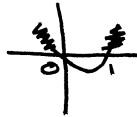
$$= 20k^2 - 20k$$

For real roots $\Delta \geq 0$

$$\therefore 20k^2 - 20k \geq 0$$

$$20k(k-1) \geq 0$$

$$\therefore k \leq 0, k \geq 1$$



(i) $a = 4$

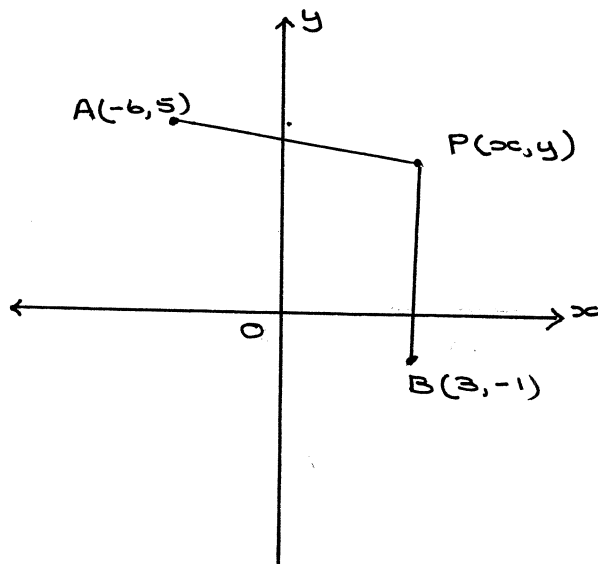
$\therefore$ directrix is $x = -11$

(ii) $(y-k)^2 = 4a(x-h)$

$$(y+2)^2 = 4 \times 4(x+7)$$

$$(y+2)^2 = 16(x+7)$$

e)



$$PA = PB$$

$$\sqrt{(x+6)^2 + (y-5)^2} = \sqrt{(x-3)^2 + (y+1)^2}$$

$$\therefore PA^2 = PB^2$$

$$(x+6)^2 + (y-5)^2 = (x-3)^2 + (y+1)^2$$

$$x^2 + 12x + 36 + y^2 - 10y + 25 = x^2 - 6x + 9 + y^2 + 2y + 1$$

$$12x + 6x - 10y - 2y + 61 - 10 = 0$$

$$18x - 12y - 51 = 0$$

$$6x - 4y - 17 = 0$$

Question 3

a) i) $\int (4x^3 + 7x^2 - 3) dx = x^4 + \frac{7}{3}x^3 - 3x + C$

ii) $\int \frac{6x^3 - 7x}{x^2} dx = \int (6x - \frac{7}{x}) dx$

$$= 3x^2 - 7 \log_e x + C$$

iii) $\int x\sqrt{x} dx = \int x^{\frac{3}{2}} dx$

$$= \frac{2}{5} x^{\frac{5}{2}} + C$$

$$= \frac{2}{5} x^2 \sqrt{x} + C$$

b) $\int_0^2 (2x-1)^3 dx = \left[\frac{(2x-1)^4}{8} \right]_0^2$

$$= \left[\frac{(2 \times 2 - 1)^4}{8} \right] - \left[\frac{(2 \times 0 - 1)^4}{8} \right]$$

$$= \frac{81}{8} - \frac{1}{8}$$

$$= 10$$

c) $\frac{dy}{dx} = 3 + 2x - x^2$

$$y = \int (3 + 2x - x^2) dx$$

$$= 3x + x^2 - \frac{1}{3}x^3 + C$$

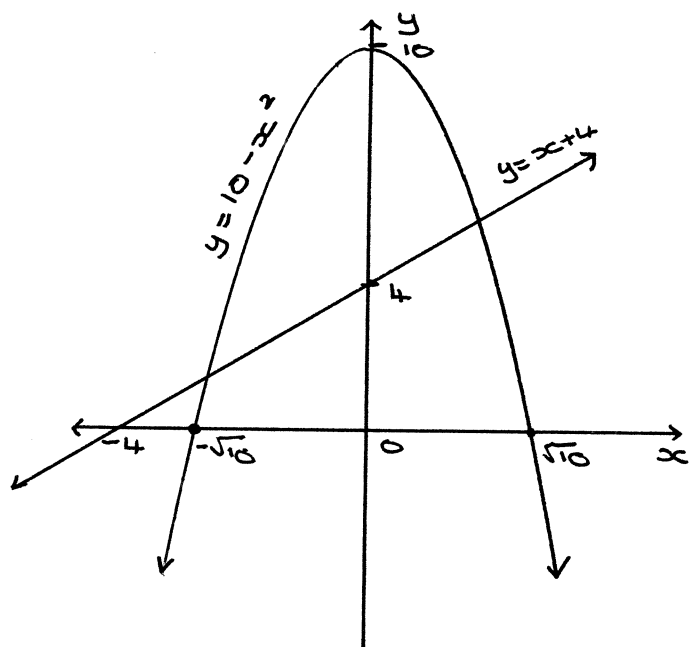
The curve passes through (2, 5)

$$\therefore 5 = 3(2) + 2^2 - \frac{1}{3}(2)^3 + C$$

$$5 = 6 + 4 - \frac{8}{3} + C$$

$$C = -\frac{7}{3}$$

$$\therefore y = -\frac{1}{3}x^3 + x^2 + 3x - \frac{7}{3}$$



$$y = 10 - x^2 \quad \text{--- ①}$$

$$y = x + 4 \quad \text{--- ②}$$

$$\textcircled{1} = \textcircled{2}$$

$$x + 4 = 10 - x^2$$

$$x^2 + x - 6 = 0$$

$$(x + 3)(x - 2) = 0$$

$$x = -3, 2$$

$$\text{when } x = -3$$

$$\text{when } x = 2$$

$$\text{sub ②}$$

$$\text{sub ②}$$

$$\begin{aligned} y &= x + 4 \\ &= -3 + 4 \\ &= 1 \end{aligned}$$

$$\begin{aligned} y &= x + 4 \\ &= 2 + 4 \\ &= 6 \end{aligned}$$

$\therefore$ points of intersection are $(-3, 1)$ $(2, 6)$

$$A = \int_{-3}^2 (10 - x^2) dx - \int_{-3}^2 (x + 4) dx$$

$$= \int_{-3}^2 (10 - x^2 - x - 4) dx$$

$$= \int_{-3}^2 (6 - x - x^2) dx$$

$$A = \left[6x - \frac{1}{2}x^2 - \frac{1}{3}x^3 \right]_{-3}^2$$

$$A = \left[6(2) - \frac{1}{2}(2)^2 - \frac{1}{3}(2)^3 \right] - \left[6(-3) - \frac{1}{2}(-3)^2 - \frac{1}{3}(-3)^3 \right]$$

$$= \left[12 - 2 - \frac{8}{3} \right] - \left[-18 - \frac{9}{2} + 9 \right]$$

$$= \frac{22}{3} + \frac{27}{2}$$

$$= 20 \frac{5}{6}$$

$\therefore$ Area is $20 \frac{5}{6}$ units²

Question 4

a) (i) Let $y = e^{4x+8}$

$$\frac{dy}{dx} = 4e^{4x+8}$$

(ii) Let $y = \frac{e^x}{e^x + 1}$

$$= \frac{u}{v}$$

$$\frac{dy}{dx} = \frac{v \frac{du}{dx} - u \frac{dv}{dx}}{v^2}$$

$$\begin{aligned} u &= e^x & v &= e^x + 1 \\ \frac{du}{dx} &= e^x & \frac{dv}{dx} &= e^x \end{aligned}$$

$$= \frac{e^x(e^x + 1) - e^x \cdot e^x}{(e^x + 1)^2}$$

$$= \frac{e^{2x} + e^x - e^{2x}}{(e^x + 1)^2}$$

$$= \frac{e^x}{(e^x + 1)^2}$$

b) $\int_0^2 e^{3x} dx = \left[\frac{1}{3} e^{3x} \right]_0^2$

$$= \left[\frac{1}{3} e^{3(2)} \right] - \left[\frac{1}{3} e^{3(0)} \right]$$

$$= \frac{1}{3} e^6 - \frac{1}{3}$$

$$= \frac{1}{3} (e^6 - 1)$$

$$c) \int 8x^3 e^{x^4} dx = 2 \int 4x^3 e^{x^4} dx \\ = 2e^{x^4} + C$$

$$d) y = e^{x^2-1}$$

$$\frac{dy}{dx} = 2x(e^{x^2-1}) \text{ at } x=1 \\ = 2(e^{1^2-1}) \\ = 2$$

$$\text{when } x=1, y = e^{x^2-1} \\ = e^{1-1} \\ = 1$$

$$\therefore \text{ point } (1, 1) \quad \frac{dy}{dx} = 2$$

$$y - y_1 = m(x - x_1)$$

$$y - 1 = 2(x - 1)$$

$$\therefore y = 2x - 1$$

$$e) y = e^x - e^{-x} \\ y^2 = (e^x - e^{-x})^2 \\ = e^{2x} - 2 + e^{-2x}$$

$$V = \pi \int_0^b y^2 dx$$

$$= \pi \int_0^{\frac{1}{2}} (e^{2x} - 2 + e^{-2x}) dx$$

$$= \pi \left[\frac{1}{2} e^{2x} - 2x - \frac{1}{2} e^{-2x} \right]_0^{\frac{1}{2}}$$

$$= \pi \left[\frac{1}{2} e^{2(\frac{1}{2})} - 2(\frac{1}{2}) - \frac{1}{2} e^{-2(\frac{1}{2})} \right] - \pi \left[\frac{1}{2} e^0 - 0 - \frac{1}{2} e^0 \right]$$

$$= \pi \left[\frac{1}{2} e - 1 - \frac{1}{2} e^{-1} \right]$$

$$= \frac{\pi}{2} \left[e - 2 - \frac{1}{e} \right]$$

$$\therefore \text{Volume is } \frac{\pi}{2} (e - 2 - \frac{1}{e}) \text{ units}^3$$

x	0	1	2	3	4
$f(x) = e^{-x^2}$	1	e^{-1}	e^{-4}	e^{-9}	e^{-16}

$$\int_a^b f(x) dx \approx \frac{b-a}{6} \left[f(a) + 4f\left(\frac{a+b}{2}\right) + f(b) \right]$$

$$\int_0^4 e^{-x^2} dx \approx \frac{4-0}{6} \left[1 + 4(e^{-1} + e^{-9}) + 2e^{-4} + e^{-16} \right]$$

$$\approx 0.8362142647$$

$$= 0.84 \text{ to 2 dp.}$$

Questions

$$a) (i) y = \log_e(2x+1)$$

$$\frac{dy}{dx} = \frac{2}{2x+1}$$

$$(ii) y = \log_e(2x+1)(x-5)$$

$$= \log_e(2x^2 - 9x - 5)$$

$$\frac{dy}{dx} = \frac{4x-9}{2x^2-9x-5}$$

$$(iii) y = x^3 \log_e x$$

$$= uv$$

$$\frac{dy}{dx} = v \frac{du}{dx} + u \frac{dv}{dx}$$

$$= \log_e x \times 3x^2 + x^3 \times \frac{1}{x}$$

$$= 3x^2 \log_e x + x^2$$

$$= x^2(3 \log_e x + 1)$$

$$b) \int_2^3 \frac{8x}{2x^2+7} dx = 2 \int_2^3 \frac{4x}{2x^2+7} dx$$

$$= \left[2 \log_e(2x^2+7) \right]_2^3$$

$$= 2 \log_e(2 \times 3^2 + 7) - 2 \log_e(2 \times 2^2 + 7)$$

$$= 2 \log_e 25 - 2 \log_e 15$$

$$= 2 \log_e \left(\frac{25}{15} \right)$$

$$= 2 \log_e \left(\frac{5}{3} \right) = \log_e \left(\frac{25}{9} \right)$$

$$c) i) \frac{4x+3}{2x+1} = 2 + \frac{1}{2x+1}$$

$$\text{L.H.S} = 2 + \frac{1}{2x+1}$$

$$= \frac{2(2x+1) + 1}{2x+1}$$

$$= \frac{4x+2+1}{2x+1}$$

$$= \frac{4x+3}{2x+1}$$

$$= \text{R.H.S}$$

$$\therefore \frac{4x+3}{2x+1} = 2 + \frac{1}{2x+1}$$

$$\begin{aligned} ii) \int \frac{4x+3}{2x+1} dx &= \int 2 + \frac{1}{2x+1} dx \\ &= \int 2 dx + \frac{1}{2} \int \frac{2x}{2x+1} dx \\ &= 2x + \frac{1}{2} \log_e(2x+1) + c \end{aligned}$$

$$d) 2\log_3 6 + \log_3 18 - 3\log_3 2$$

$$= \log_3 6^2 + \log_3 18 - \log_3 2^3$$

$$= \log_3 \left(\frac{36 \times 18}{8} \right)$$

$$= \log_3 81$$

$$= \log_3 3^4$$

$$= 4\log_3 3$$

$$= 4$$

$$e) i) \log_2 64 = x$$

$$2^x = 64$$

$$2^x = 2^6$$

$$x = 6$$

$$ii) 3^x = 5$$

$$\log_e 3^x = \log_e 5$$

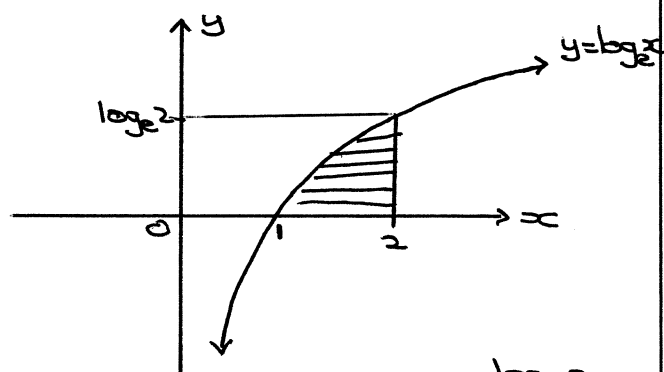
$$x \log_e 3 = \log_e 5$$

$$x = \frac{\log_e 5}{\log_e 3}$$

$$\approx 1.464973521$$

$$= 1.46 \text{ to 2dp}$$

f)



$$A = \text{Area rectangle} - \int_0^{\log_e 2} e^y dy$$

$$= 2 \times \log_e 2 - \{ [e^{\log_e 2} - e^0] \}$$

$$= 2\log_e 2 - \{2 - 1\}$$

$$= 2\log_e 2 - 1$$

$$\therefore \text{Area is } (2\log_e 2 - 1) \text{ units}^2$$